Ergonomic Risk Assessment Using Human Pose Estimation with MediaPipe Pose

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Abstract

This research utilizes Human Pose Estimation (HPE) to detect improper working postures that may lead to Work-Related Musculoskeletal Disorders. These postures are evaluated according to ergonomic principles using the OWAS method, which assesses the back, arms, legs, and the weight of the load during the task with codes indicating risk categories. In the experiment, both Two-dimensional (2D) and three-dimensional (3D) keypoints of body joints obtained from MediaPipe Pose and the angles of the joints relative to 72 different postures, encompassing all combinations of the code for the three body parts, were used as features for setting the criteria in posture prediction. The best classifier obtained from the experiment provided accuracy rates of 90.74% for the back, 92.71% for the arms, and 85.53% for the legs, respectively. When the program was applied to evaluate working postures in a simulated packing station, the accuracy rates for the back, arms, and legs were 70.42%, 82.78%, and 71.25%, respectively. The accuracy in identifying the risk category, which considers all body parts simultaneously, is 77.64%.

CCS Concepts

• **Computing methodologies**; • **Machine learning; Machine learning approaches; Rule learning.**;

Keywords

ergonomics, OWAS, human pose estimation, MediaPipe Pose

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1 Introduction

In today's industrial and occupational settings, the health and safety of workers are critical concerns. Work-Related Musculoskeletal Disorders (WMSDs) are among the most common occupational injuries, often resulting from poor working postures, repetitive movements, and excessive physical loads. These disorders not only affect the well-being of employees but also lead to significant economic costs due to lost productivity and increased healthcare expenses. Effective ergonomic risk assessment is essential to identify and mitigate the factors contributing to WMSDs, ensuring a safer and more productive work environment. The Ovako Working Posture Assessment System (OWAS) is a widely recognized method for evaluating working postures and their associated risks. OWAS, which was developed in the late 1970s, systematically categorizes postures based on the positions of the back, arms, and legs, as well as the load being handled. Each posture-load combination is assigned a specific code and classified into one of four action categories, ranging from no corrective measures needed to immediate intervention required. This approach allows for a detailed analysis of ergonomic risks and prioritizes necessary corrective actions.

In traditional risk assessments, the evaluation is conducted through direct observation of working postures or through questionnaires. These methods may be inaccurate and inefficient due to subjective bias. With the advent of advanced computer vision technologies, Human Pose Estimation (HPE) has emerged as a powerful tool for automatically detecting and tracking key points of the human body, such as joints, to infer body postures in images or videos. By utilizing deep learning and neural networks, these pose estimation algorithms achieve a high degree of accuracy and operate in real-time. This enables continuous monitoring of workers in a non-intrusive manner. Integrating HPE with OWAS presents a significant advancement in ergonomic risk assessment. This innovative approach combines the systematic posture classification of OWAS with the automated data collection capabilities of pose estimation. By continuously capturing real-time posture data, it becomes possible to identify and detect hazardous postures more accurately and frequently, thus enhancing the effectiveness of ergonomic assessment.

In this research, HPE with MediaPipe Pose will be used to develop an automated OWAS ergonomic risk assessment program. The operation of the program is divided into three parts: image acquisition, risk assessment, and reporting. The image acquisition part of the program involves uploading images and capturing real-time images through a webcam. In the risk assessment part, a rule-based classifier will be developed to identify posture codes for the back, arms, and legs. In the reporting part, the program will report the risk score for each body part and the risk category, along with the interpretation of the results. This program will be tested

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Table 1: OWAS code for body part and load

Part	Code	Posture Description
Back	1	Back Straight
	2	Back bent forward or backward
	3	Back bent sideways or twisted
	4	Back bent and twisted or Back bent forward or backward and bent sideways
Arms	1	Both arms below shoulder level
	2	One arm at or above shoulder level
	3	Both arms at or above shoulder level
Legs	1	Sitting, with legs positioned below seat level.
	2	Standing on both legs straight
	3	Standing on one leg, which is the straight leg
	4	Standing on both legs bent
	5	Standing on one leg, which is the bent leg
	6	Kneeling on one or both knees
	7	Walking
Load	1	No load or less than 10 kg.
	2	Load of 10 kg. or more but less than 20 kg.
	3	Load of 20 kg. or more

in a simulated packing station to be used as a teaching aid and for training employees. The remainder of this paper will be structured as follows: Section 2 will review related research on ergonomic risk assessment using the OWAS method and HPE. The subsequent section will describe the research methodology. The experimental results and their analysis will be presented in Section 4, and the findings will be summarized in Section 5.

2 LITERATURE REVIEW

2.1 Ergonomic Risk Assessment using OWAS Method

Ergonomic risk assessment is crucial for identifying and mitigating potential hazards in the workplace that can lead to Work-Related Musculoskeletal Disorders (WMSDs). Consequently, various ergonomic assessment tools have been proposed to monitor and control postural ergonomic risks. Examples include the Rapid Upper Limb Assessment (RULA), which focuses on upper limbs; the Rapid Entire Body Assessment (REBA), which assesses ergonomic risks in tasks involving whole-body movement; the Ovako Working Posture Assessment System (OWAS), which identifies and assesses working postures; ISO 11226:2000, which establishes acceptable angles and holding times for working postures; and EN 1005-4, which provides guidelines for evaluating machine-related postures and body movements in machinery design. OWAS is designed to cover general postures, offering a straightforward and practical approach that is well-documented and applicable to personnel across various fields such as engineering, industry, healthcare, and agriculture, without requiring specialized training [3]. The traditional OWAS assessment procedure involves observing the worker's postures during their tasks. This includes noting the positions of the back, arms, and legs, as well as the amount of force exerted. Each posture is then assigned a code based on the positions of the back, arms, and legs, and the amount of load, as detailed in Table 1. Accordingly, each of individual posture during the task has 4-digit OWAS code. Accordingly, each individual posture during the task has a 4-digit OWAS code. The 4-digit codes are classified into four risk categories, as shown in Table 2.

2.2 Human Pose Estimation

Human pose estimation (HPE) involves detecting and tracking keypoints of the human body in images or video frames to determine body postures. This technology has significant applications in various fields, including sports analytics, healthcare, virtual reality, and ergonomics. Human pose models can be classified along several axes, such as: type of input modality (Red-Green-Blue (RGB) image, depth image, infrared image); number of cameras (single-view or multi-view); human body models (kinematic, planar, or volumetric); type of images (static images or frames from video sequences); number of people being tracked (single-person or multi-person pose estimation); and dimensionality (2D or 3D pose estimation). The combination of classifications across different dimensions leads to the development of various HPE models tailored for specific applications and requirements [2]. In this research, HPE is utilized by capturing RGB images through a single camera (single view) to estimate a skeleton-based human body model. The system continuously receives real-time images of a single worker (single-person pose estimation) to analyze postures in 3D space.

The 3D pose estimation involves determining the articulated 3D joint (keypoint) regions of the human body from an image or video [1]. It calculates the 3D posture from an RGB image. The purpose of 3D HPE is to use an image of a person to determine the XYZ coordinates of a particular set of key points on the human body. Following the extraction of joint positions, the activity assessment model examines a person's posture and analyzes the person's real motion in a series of frames from a video stream. Generally, it is

Table 2: Risk categories of OWAS

Categories	Posture Description	Required Action
1	Postures have no detrimental effects.	No action required.
2	Postures have some detrimental effects.	Corrective actions required in near future.
3	Postures have a clear detrimental effect.	Corrective actions required as soon as possible.
4	Postures have a considerable detrimental effect.	Immediate corrective actions required.

more challenging to recover 3D poses from 2D RGB images due to increased ambiguities in estimation, and it requires more storage compared to 2D images [6]. The earliest approaches for 3D HPE from single images were typically based on discriminative methods, treating pose estimation as either a regression or classification problem. A mapping function was learned from a space of image features or computed based on body part information [10] to estimate the pose in 3D space. The exceptional results of deep learning methods in computer vision have made it a common trend to use deep networks to automatically learn key information from images. Some research relies on supervised learning to directly regress joint locations and predict the 3D pose from 2D monocular images without intermediate supervision [4, 5, 11, 12, 17, 18]. In this case, models must be trained using a 3D pose-annotated image dataset. This can limit the models' ability to generalize well to images from different environments, such as those encountered in real-world scenarios. To address this issue, some approaches use both 2D datasets from the wild and 3D motion capture (MoCap) datasets to guide and improve training. Monocular view is a critical aspect of 3D HPE from monocular images, which only provide a single 2D view of the scene. As a result, algorithms must rely solely on the information contained in that single image to estimate the 3D pose of the human subject. This can be challenging, as the loss of depth information can make accurate pose estimation more difficult. However, there are advantages to using monocular images. For instance, monocular images are often easier to acquire and more widely used in real-world scenarios, as they do not require specialized equipment or setups.

Open-source human pose estimation frameworks, such as OpenPose, AlphaPose, and MediaPipe Pose, have democratized access to advanced computer vision techniques, enabling a wide range of applications and research advancements. Each framework has its strengths. MediaPipe Pose, for instance, is advantageous for cross-platform applications. It is designed for real-time pose tracking and supports multiple platforms, including Android, iOS, and the web. MediaPipe is an open-source framework developed by Google that provides a robust platform for building multimodal machine learning pipelines. It is designed to facilitate the development of real-time machine learning applications by providing pre-built solutions for various tasks, such as pose estimation with MediaPipe Pose, hand sign language recognition with MediaPipe Hand [9], face detection and mesh with MediaPipe Face, and more. The MediaPipe Pose estimation methodology performs 3D pose estimation by lifting 2D keypoints detected in images or video frames into 3D space. This process consists of two main stages: 2D Pose Detection, which involves the initial detection of 2D keypoints using convolutional neural networks (CNNs), and 3D Lifting, where a model predicts the depth (z-coordinate) for each 2D keypoint to obtain 3D coordinates. MediaPipe Pose utilizes a variety of datasets to train its models, including: COCO (Common Objects in Context), MPII Human Pose Dataset, LSP (Leeds Sports Pose) Dataset, Human3.6M. These datasets collectively help MediaPipe Pose achieve robust performance across a variety of conditions and poses. The MediaPipe pose estimation methodology consists of two main stages. The first stage, 2D Pose Detection, involves the initial detection of 2D keypoints using convolutional neural networks (CNNs). The second stage, 3D Lifting, predicts the depth (z-coordinate) for each 2D keypoint to obtain 3D coordinates. The pose landmarker model of MediaPipe Pose tracks 33 body landmark locations, representing the approximate positions of various body parts, as shown in Figure 1.

2.3 Integration of Human Pose Estimation and Ergonomic Risk Assessment

The integration of Human Pose Estimation (HPE) with ergonomic risk assessment has been applied in various industries such as manufacturing and assembly lines, healthcare and nursing, office environments, and construction. This involves using open-source software tools for 3D HPE to implement ergonomic risk assessment methods like OWAS, RULA, and REBA. This approach was developed to improve the efficiency of traditional risk assessments by allowing for automatic evaluation scores from images, videos, or in real-time, and by reducing the bias of evaluators who require direct observation of employees.

From the literature review, it was found that HPE is mostly applied with the RULA or REBA methods for risk assessment. In contrast, OWAS, which can assess risk in a wide variety of tasks, has not been extensively studied. The input of keypoint data in HPE for risk assessment using the OWAS method is often studied with Kinect or MoCap systems rather than commercially available RGB cameras [14]. However, Kinect is vulnerable to outdoor conditions where solar IR and ferromagnetic radiation can cause significant noise or even wash out the scene generated by the Kinect. Additionally, Kinect has higher power consumption, lower resolution, and is not as widely available or inexpensive as ordinary cameras [16].

Therefore, [8] presents a computer vision-based method to perform ergonomic risk assessment using off-the-shelf RGB cameras. This method takes snapshots or digital videos as input, determines body keypoints with OpenPose, and automatically calculates RULA scores. The computed RULA scores closely align with the assessments of seven specialists in the field. [7] developed an OpenPose-based system for computing joint angles and RULA/REBA scores and validated it against a reference MoCap system, comparing its

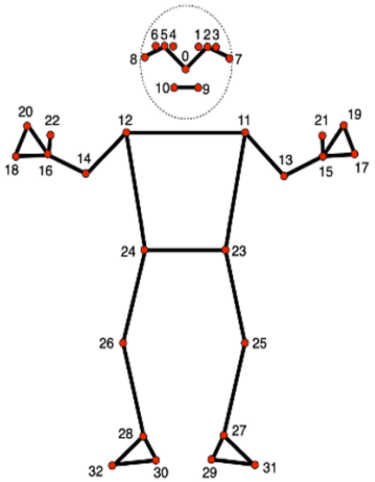

Figure 1: Body landmark locations [19]

Table 3: Keypoints and features for body part classification

Part	Code	Keypoints	Features	2D/3D coordinate
Back	1, 2	11,12,23,24	Middle body line angle	2D
	3, 4	11,12, 23,24	Middle body line angle	3D
		11,12	Shoulder line angle	
		23,24	Hip line angle	
Arms	1 - 3	11,13,23	Left shoulder angle	2D
		12,14, 24	Right shoulder angle	
Legs	1 - 6	23,25	Left knee-hip line angle	2D
		24,26	Right knee-hip line angle	
		23,25, 27	Left knee angle	
		24,26, 28	Right knee angle	
		25	Left knee keypoint	
		26	Right knee keypoint	

performance to a Kinect-based system. The findings suggested that OpenPose could be a promising technology for measuring joint angles and conducting semi-automatic ergonomic postural assessments in real-world workspaces, where conditions are often non-ideal. The study in [15] applies a 2D HPE framework to automatically calculate OWAS scores. This study aims to develop an ergonomic posture recognition technique based on a 2D ordinary camera for construction hazard prevention through view-invariant features in 2D skeleton motion. Three posture classifiers for the human back, arms, and legs were employed to ensure that three postures could be recognized simultaneously in one video frame. The overall accuracies tested with three new activities regarding ergonomic assessment scores captured from different camera heights and viewpoints were 89.2%, 88.3%, and 87.6%, respectively.

3 RESEARCH METHODOLOGY

The research methodology consists of five main steps as follows:

- Study of posture codes used in the OWAS method. This step examines body postures segmented according to the OWAS method into three parts: back, arms, and legs. Each part is categorized into 4, 3, and 7 codes, respectively, as shown in Table 1. Note that for the legs, the codes do not include code 7: Walking.
- Posture Classification. In this step, a rule-based classifier will be developed to identify the relationships between features associated with different postures and the types of postures for each body part to assess ergonomic risks. The features include keypoints (the XYZ coordinates provided by MediaPipe Pose, as shown in Figure 1), the angles between joints, and the relevant body part connection lines. These features will be used to classify the posture of the back, arms, and legs, as summarized in Table 3.

According to the literature review, while 3D keypoints provide information on the z-axis, predicting accurate values remains challenging. Therefore, this research will experiment with both 2D and 3D keypoints. The column "2D/3D coordinate" refers to selecting coordinates from different axes for angle calculation. In 2D, XY coordinates are used, while in 3D, XYZ coordinates are utilized.

The values entered in Table 3 will be the types of keypoints that yield the most accurate posture prediction results. When setting the parameters for posture classification, the angle values are specified according to the OWAS method [13] used in direct assessment. These values will be adjusted to compensate the image distortion using offset values. The offset will be tuned within a range of plus or minus 5%, tested at intervals of 1%. In this step, 72 working postures will be used, with 12 images per posture, totaling 864 images, for tuning the offset. The classifier model that achieves the highest accuracy will be used in the program for automatically assessing ergonomic risk using the OWAS method.

- Testing the posture classifier. The developed posture classifier will be tested by evaluating working postures in a simulated packing station. The activities will include packing goods into boxes, arranging goods into crates, and lifting goods onto shelves. Two participants will be involved in the testing, covering 72 working postures. A camera will be set up to capture full-body images of the participants at a 45-degree angle to their front. Ten images per posture will be collected at the simulated workstation. The accuracy, recall, and precision metrics described in step 4 will be used to evaluate the program.
- Classifier model evaluation. The posture codes predicted by the program will be compared with assessments made by an ergonomics expert. A multi-class confusion matrix for each body part will be constructed. These multi-class confusion matrixes will be converted into one-vs-all type matrixes (binary-class confusion matrixes) to calculate class-wise metrics such as accuracy, precision, and recall.
- The development of an automated ergonomic assessment program using the OWAS method. The program comprises three main parts: data input, OWAS assessment using a developed classifier model, and output display.

For the input data, the program can receive real-time data via webcam or images. The load level that participants work with is input into the program at the beginning. If there are loads of different levels, the work is divided into smaller task elements before analysis.

In the OWAS assessment part, the program evaluates the risk of postures based on the input data using the developed posture classifier. It analyzes the postures and assesses the risk level for three body parts: the back, arms, and legs. The scores are then sent to the output display section.

In the output display part, the program presents the assessment scores for the three body parts, along with the risk category, on the screen. The results are displayed in a clear and user-friendly manner, allowing users to easily understand the risk categories and take necessary actions.

4 RESULTS AND DISCUSSION

The research results are divided into two parts: the evaluation of a classifier in assessing ergonomic risks using the OWAS method, and the application of the program.

4.1 The Evaluation Results of a Classifier

During the process of tuning the features and offsets in the criteria for classifying all 72 posture codes using 864 images, it was found that the program predicted postures and assessed posture scores, summarized in confusion matrixes for the back, arms, and legs, as shown in Figure 2. The rightmost column shows the precision for each body part code, while the last row shows the recall. The accuracy of each body part code is indicated in the diagonal cells, and the bottom-right cell displays the overall accuracy. According to the confusion matrixes, the highest accuracy was achieved for the arm section at 92.71%, followed by the back at 90.74% and the legs at 85.53%. When analyzing postures with a recall value below 90.00%, it was found that issues arise in the back section, specifically for postures requiring information along the z-axis. For back codes 2 and 4, which involve bending forward (forming an angle with the z-axis) and twisting (around the z-axis) while bending, these postures were the most frequently misclassified by the classifier. In the leg section, the classifier often confused code 1 (sitting on a chair) with code 4 (standing with both legs bent or squatting) and code 3 with code 5. Both codes 3 and 5 involve one leg being bent and the other straight, but they differ in weight distribution: code 3 places the weight on the straight leg, while code 5 places the weight on the bent leg. Although the program was tested using 3D keypoints for these postures, it was found that using 2D keypoints or calculating the angles of the relevant joints from 2D keypoints yielded better results. Further diagnosis revealed that the positions of the keypoints on the image did not align with the specified joint positions due to self-occlusion and the error in evaluating the z-coordinate.

The classifier testing involved using images of operations at a simulated packing station, totaling 72 postures, with 10 images per posture. The accuracy in predicting the postures for the back, arms, and legs was 70.42%, 82.78%, and 71.25%, respectively, which is a decrease from the parameter tuning stage. Analysis of the test results revealed that the postures the classifier confused were the same as those during the parameter tuning stage. The decrease in accuracy was attributed not only to self-occlusion but also to inter-object occlusion, where parts and crates used in packaging obscured parts of the body, making evaluation more challenging.

When considering the risk categories obtained from the posture assessment, which takes into account the back, legs, and arms simultaneously, it was found that the program's accuracy in identifying the risk categories is 77.64%. If we consider categorizing the risk categories based on the required action for working postures related to occupational injuries, they can be divided into two groups: Group 1 consists of risk categories 1 and 2, which do not require immediate improvement in working posture, and Group 2 consists of risk categories 3 and 4, which require immediate improvement due to their impact on occupational injury. Based on the experimental results, it was found that identifying a risk category from Group 1 as Group 2, known as a false alarm, accounts for 11.46% and will lead to unnecessary work improvements, causing additional costs and time. On the other hand, identifying a risk category from Group 2 as Group 1, known as a missed alarm, means that work that should be improved is not addressed, resulting in ergonomic risks for employees, accounting for 9.35%.

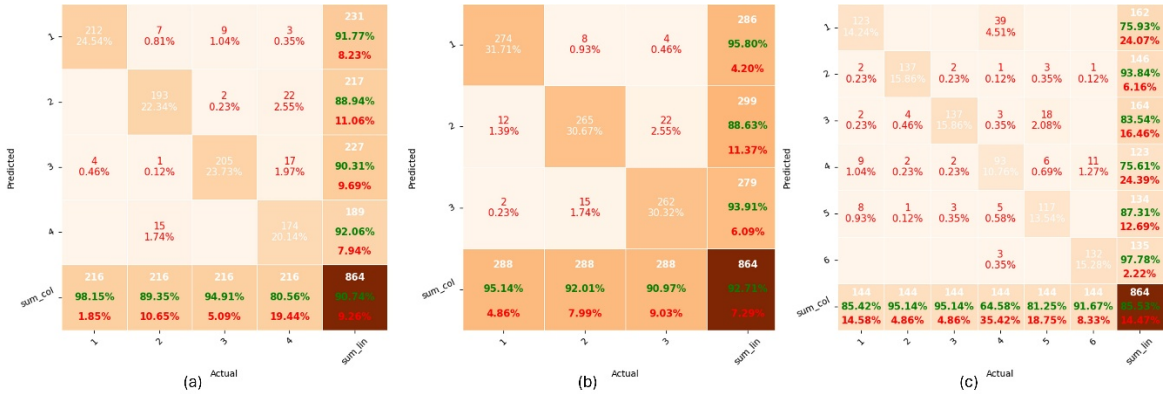

Figure 2: Confusion matrixes for (a) back, (b) arms, and (c) legs.

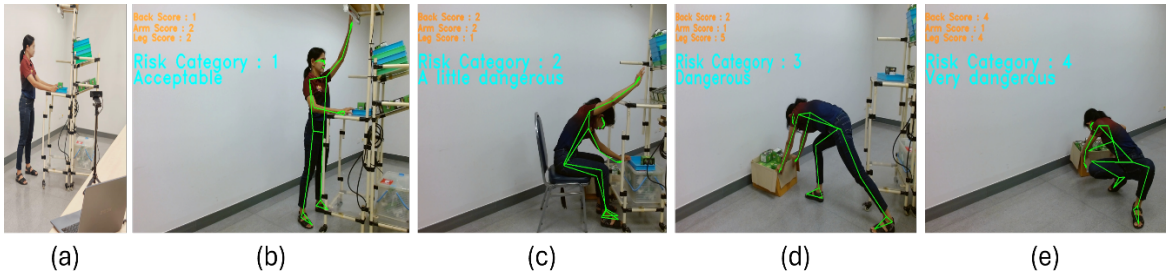

Figure 3: (a) setting up the equipment, (b) – (e) reporting the results on the screen

4.2 The Application of The Program

The program for automatic risk assessment using the OWAS method is designed to accept real-time input through a webcam or by reading images from a file. For real-time image capture, the webcam should be set up at a 45-degree angle to the operator's body. The reporting section then provides the code for each body part, the risk category, and its interpretation, as shown in Figure 3.

When posture assessment from images is needed, simply upload the images into a specified folder. The program will then display the assessment results for each part and the risk category on the screen. For evaluating a large number of images, the program can process them and save the results for each image in an Excel file.

5 CONCLUSION

This research applies HPE for ergonomic risk assessment by using keypoints from MediaPipe Pose in a rule-based classifier to categorize postures according to risk assessment codes for different body parts using the OWAS method. Experimental results using both 2D and 3D keypoints show that 2D keypoints provide higher accuracy than 3D keypoints for postures identifiable in both 2D and 3D planes. For example, in the arm code, determining the angle of the line passing through the keypoints of the elbow, arm, and hip is clearer when calculated in the 2D plane. Conversely, 3D keypoints offer better results than 2D keypoints in conditions where avoiding z-coordinate data is not possible, such as with back codes 3 and 4, which involve rotation around the z-axis. Projecting the posture onto a 2D plane makes the line connecting the torso and hips unclear, preventing accurate distinction between these postures. Therefore, for back codes 3 and 4, the use of 3D keypoints yields better results than 2D keypoints.

From the experimental results, it can be concluded that the rule-based algorithm developed using features related to postures in each code, along with offset tuning to compensate for occlusion and errors in keypoint estimation, can accurately predict postures of the back, arms, and legs at an acceptable level. When applying this rule-based classifier to work postures in a simulated packing station, the accuracy rates were found to be 70.42%, 82.78%, and 71.25%, respectively. Considering the risk categories obtained from the posture assessment, which takes into account the back, legs, and arms simultaneously, the program's accuracy in identifying the risk category was 77.64%.

The developed classifier will be incorporated into the risk assessment component of the automated OWAS ergonomics risk assessment program. The program can receive images for risk assessment through both upload and real-time image capture. The display of risk categories and their meanings on the screen allows users to clearly understand the assessment results. This program can be used as a teaching aid in the Ergonomics course, an important field for industrial engineers, and can be further developed for use in employee training or real-world applications.

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